

Simple Storage Service (S3)

What is an S3 Bucket?

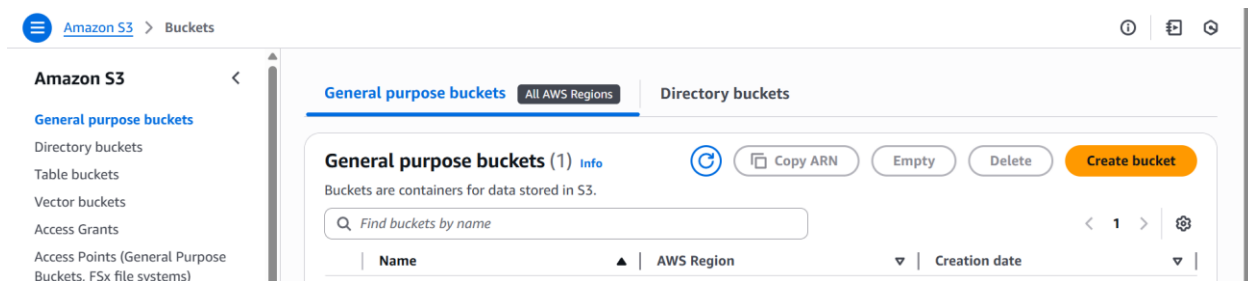
- An S3 bucket is like a main folder or container in Amazon S3 (Simple Storage Service) where you can store, organize, and manage files (called objects) in the cloud.
- *By default, you can create up to 10,000 S3 buckets per AWS account (maximum limit is 1 million buckets upon request). You can store unlimited data in S3, and each object can be up to 5 TB in size.*
- In the AWS Free Tier, you can store up to 5 GB total of data in Amazon S3 Standard for free.

The Multipart Upload feature is used when you upload a single object larger than 5 GB it's not related to the Free Tier limit, but to the size of the file being uploaded.

Bucket Creation:

- Amazon S3 allows people to store objects (files) in “buckets” (directories)
- Buckets must have a globally unique name (across all regions all accounts)
- Buckets are defined at the region level
- S3 looks like a global service but buckets are created in a region

Click on create bucket



Amazon S3 > Buckets > Create bucket

Create bucket

info

Buckets are containers for data stored in S3.

General configuration

AWS Region

Asia Pacific (Mumbai) ap-south-1

Bucket type

Info

☒ General purpose

Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory

Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name

Info

mentore-solution001

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Keep Default and click on create Bucket

Object:

Here you can upload your data. 1 Root user can create 100 Buckets and in each bucket we can store data up to 5TB

Amazon S3 > Buckets > mentore-solution001

mentore-solution001

info

Objects

Properties

Permissions

Metrics

Management

Access Points

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

Name	Type	Last modified	Size	Storage class
No objects				
You don't have any objects in this bucket.				

Upload

Amazon S3 > Buckets > mentore-solution001 > Upload

Upload

info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (1 total, 3.3 MB)

Remove

Add files

Add folder

All files and folders in this table will be uploaded.

Find by name

< 1 >

Name	Folder	Type	Size
AWS basics.pdf	-	application/pdf	3.3 MB

Destination

info

Destination

Destination [Info](#)

Destination
[s3://mentore-solution001](#)

► Destination details
 Bucket settings that impact new objects stored in the specified destination.

▼ Permissions
 Grant public access and access to other AWS accounts.

ⓘ This bucket has the [bucket owner enforced](#) setting applied for Object Ownership. Use bucket policies to control access. [Learn more](#)

▼ Properties
 Specify storage class, encryption settings, tags, and more.

So, there are 8 types of storage classes.

1. **S3 Standard** → For **frequently accessed** data. Fast, but costlier.
2. **S3 Intelligent-Tiering** → **Auto-moves data** between frequent & infrequent tiers.
3. **S3 Standard-IA (Infrequent Access)** → For data you **don't use daily**, but need quickly when required.
4. **S3 One Zone-IA** → Same as IA but stored in **one AZ only** → cheaper, less durable.
5. **S3 Glacier Instant Retrieval** → Rarely accessed, but need in **seconds**.
6. **S3 Glacier Flexible Retrieval** → Archive, retrieval in **minutes to hours**.
7. **S3 Glacier Deep Archive** → Cheapest, for **long-term compliance**, retrieval takes **12–48 hrs**.
8. **S3 Outposts** → S3 storage **on-premises** (local data center).

▼ Properties
 Specify storage class, encryption settings, tags, and more.

Storage class [Info](#)
 Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)

	Storage class	Designed for	Bucket type	Availability Zones	Min storage duration	Min billable object size	Monitoring and auto-tiering fees	Retrieval fees
☒	Standard	Frequently accessed data (more than once a month) with milliseconds access	General purpose	≥ 3	-	-	-	-
☐	Intelligent-Tiering	Data with changing or unknown access patterns	General purpose	≥ 3	-	-	Per-object fees apply for objects ≥ 128 KB	-
☐	Standard-IA	Infrequently accessed data (once a month) with milliseconds access	General purpose	≥ 3	30 days	128 KB	-	Per-GB fees apply
☐	One Zone-IA	Recreatable, infrequently accessed data	General purpose or	1	30 days	128 KB	-	Per-GB fees

This just means **your files are encrypted automatically while stored in S3**, so even if someone gets access to the bucket, they can't read the files.

Server-side encryption [Info](#)

Server-side encryption protects data at rest.

Server-side encryption

- ☐ Don't specify an encryption key
The bucket settings for default encryption are used to encrypt objects when storing them in Amazon S3.
- ☒ Specify an encryption key
The specified encryption key is used to encrypt objects before storing them in Amazon S3.

Encryption settings [Info](#)

- ☐ Use bucket settings for default encryption
- ☒ Override bucket settings for default encryption

Encryption type [Info](#)

Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the [Amazon S3 pricing page](#).

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Checksums are like digital fingerprints for data files. When you store or transfer a file in AWS S3, a checksum is created based on the file's content

Checksums

Checksums are used for data integrity verification of new objects. [Learn more](#)

Checksum function

Checksum functions are used to calculate the checksum value. For objects smaller than 16 MB, only the full object checksum type is supported, for all checksum algorithms.

CRC64NVME (recommended)

Precalculated value - optional

When you provide a precalculated value for a single object, S3 compares it to the value it calculates using the selected checksum function. If the values don't match, the upload will fail. [Learn more](#)

Enter value

The precalculated value must be a Base64 encoded string. It must not exceed 128 characters, and can contain only letters (a-z, A-Z), numbers (0-9), forward slash (/), plus (+), or equals (=).

Tags in AWS S3 are like labels you put on files (objects) to help organize and find them easily.

Tags - optional

You can use object tags to analyze, manage, and specify permissions for objects. [Learn more](#)

Key

Project

Value - optional

test

Remove

Add tag

Metadata is information about the file itself, like its size, date created, or type. In S3, metadata can be custom data that you attach to objects to describe or manage them better.

Metadata - optional

Metadata is optional information provided as a name-value (key-value) pair. [Learn more](#)

Type

User defined

Key

x-amz-meta-01

Value

testing

Remove

Add metadata

Cancel

Upload

Click upload

Upload succeeded
For more information, see the [Files and folders](#) table.

Upload: status Close

After you navigate away from this page, the following information is no longer available.

Summary		
Destination s3://mentore-solution001	Succeeded ✔ 1 file, 3.3 MB (100.00%)	Failed ✖ 0 files, 0 B (0%)

Files and folders | Configuration

Files and folders (1 total, 3.3 MB)

Properties:

keep multiple versions of the same file (object) in one bucket. When versioning is turned on, every time you change or delete a file, S3 saves the old version instead of permanently replacing or removing it.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures.

Once you enable then you can't disable it.

Edit Bucket Versioning [Info](#)

Bucket Versioning

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Bucket Versioning

☒ Suspend

This suspends the creation of object versions for all operations but preserves any existing object versions.

☐ Enable

Multi-factor authentication (MFA) delete

An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#) 

Disabled

[Cancel](#)

[Save changes](#)

Default Encryption

What it is:

- This automatically **encrypts all new objects** uploaded to your bucket.
- Even if someone forgets to turn on encryption during upload — AWS does it for you.

Why it's important:

- Protects data “at rest” (when stored).
- Prevents unauthorized users from reading your files, even if they somehow get access.

Types of encryptions:

- **SSE-S3:** Managed by AWS (default, simple).
- **SSE-KMS:** Managed by AWS Key Management Service (you control the keys).
- **SSE-C:** Customer provides their own encryption key.

Amazon S3 > Buckets > mentore-solution001 > Edit default encryption

- Directory buckets
- Table buckets
- Vector buckets
- Access Grants
- Access Points (General Purpose Buckets, FSx file systems)
- Access Points (Directory Buckets)
- Object Lambda Access Points
- Multi-Region Access Points
- Batch Operations
- IAM Access Analyzer for S3

Block Public Access settings for this account

▼ Storage Lens

Dashboards

Edit default encryption Info

Default encryption

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type Info

Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the [Storage](#) tab of the [Amazon S3 pricing page](#).

- ☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)
- ☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)
- ☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

- ☐ Disable
- ☒ Enable

Cancel Save changes

Intelligent-Tiering Archive Configurations

What it is:

- Works with **S3 Intelligent-Tiering** storage class.
- It automatically moves objects between access tiers (Frequent, Infrequent, Archive, Deep Archive) based on how often they're used.

Why it's important:

- Saves money — you pay less for files you rarely use.
- No manual management needed — AWS watches file usage and moves them automatically.

Object Lock

Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. [Learn more](#)

- ☐ Disable
- ☒ Enable

Permanently allows objects in this bucket to be locked. Additional Object Lock configuration is required in bucket details after bucket creation to protect objects in this bucket from being deleted or overwritten.

⚠ **Enabling Object Lock will permanently allow objects in this bucket to be locked**

After you enable Object Lock for a bucket, you can't disable Object Lock or suspend Versioning for that bucket. [Learn more about Using Object Lock](#)

☒ I acknowledge that enabling Object Lock will permanently allow objects in this bucket to be locked.

Default retention

Automatically protect new objects put into this bucket from being deleted or overwritten.

- ☒ Disable
- ☐ Enable

Cancel Save changes

Object Lock Enabled

- In your bucket settings, **Object Lock is enabled**.

- Object Lock uses a **WORM (Write Once Read Many)** model, which means:
 - Once an object is written, it **cannot be deleted or overwritten** until the retention period expires (or indefinitely if no period is set).

S3 Object Lock is specifically designed to prevent accidental or malicious deletion, once it's enabled, objects stay there until:

- Their **retention period** expires, OR
- You explicitly remove a legal hold (if applied), and retention rules allow deletion.

Until then, any delete operation will fail → showing Access Denied

Requester pays

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. [Learn more](#)

Requester pays
 Disabled

Edit

Requester Pays

What it is:

- Normally, the **bucket owner** pays for data download.
- But with *Requester Pays*, **the person downloading** the data pays the cost.

Why it's important:

- Useful for public datasets or shared data — when you don't want to pay others' download costs.

Edit static website hosting

Static website hosting
 Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting
☐ Disable
☒ Enable

Hosting type
☒ Host a static website
 Use the bucket endpoint as the web address. [Learn more](#)
☐ Redirect requests for an object
 Redirect requests to another bucket or domain. [Learn more](#)

ⓘ For your customers to access content at the website endpoint, you must make all your content publicly readable. To do so, you can edit the S3 Block Public Access settings for the bucket. For more information, see [Using Amazon S3 Block Public Access](#)

Index document
 Specify the home or default page of the website.

Static Website Hosting

What it is:

- Lets you host a **simple website directly from your bucket** (no need for EC2 or a web server).
- You just upload your HTML, CSS, JS files.

Permission:

Edit Block public access (bucket settings) [Info](#)

Block public access (bucket settings)

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☒ **Block public access to buckets and objects granted through *new* access control lists (ACLs)**

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☒ **Block public access to buckets and objects granted through *any* access control lists (ACLs)**

S3 will ignore all ACLs that grant public access to buckets and objects.

☒ **Block public access to buckets and objects granted through *new* public bucket or access point policies**

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☒ **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

[Cancel](#)

[Save changes](#)

Block Public Access

- This setting helps you keep your bucket private by blocking any public access (from the internet).
- Why it's important: It protects you from accidentally making data public — a very common mistake that can expose sensitive info.

Object Ownership

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☐ ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠ We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

⚠ Enabling ACLs turns off the bucket owner enforced setting for Object Ownership

Once the bucket owner enforced setting is turned off, access control lists (ACLs) and their associated permissions are restored. Access to objects that you do not own will be based on ACLs and not the bucket policy.

☒ I acknowledge that ACLs will be restored.

Object Ownership

☒ Bucket owner preferred

If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.

☐ Object writer

The object writer remains the object owner.

ℹ If you want to enforce object ownership for new objects only, your bucket policy must specify that the bucket-owner-full-control canned ACL is required for object uploads. [Learn more](#)

Object Ownership

- What it is:
Controls who owns the uploaded files (objects) inside a bucket.
- Why it's important:
Sometimes, different AWS accounts upload files to your bucket.
By default, the uploader becomes the object owner, which can cause permission issues.

Access Control List (ACL)

What it is:

An **older way** to control access to individual **objects** or **buckets**.

You can use it to **grant basic permissions** like:

- Read
- Write
- Full control

Management:

Lifecycle rule configuration

Lifecycle rule name

first

Up to 255 characters

Choose a rule scope

- ☐ Limit the scope of this rule using one or more filters
- ☒ Apply to all objects in the bucket

⚠ Apply to all objects in the bucket

If you want the rule to apply to specific objects, you must use a filter to identify those objects. Choose "Limit the scope of this rule using one or more filters". [Learn more](#)

☒ I acknowledge that this rule will apply to all objects in the bucket.

Lifecycle rule actions

Choose the actions you want this rule to perform.

- ☐ Transition current versions of objects between storage classes
This action will move current versions.
- ☐ Transition noncurrent versions of objects between storage classes
This action will move noncurrent versions.
- ☐ Expire current versions of objects
- ☐ Permanently delete noncurrent versions of objects
- ☐ Delete expired object delete markers or incomplete multipart uploads
These actions are not supported when filtering by object tags or object size.

Lifecycle Configuration

What it is: A **set of rules** that automatically **moves or deletes objects** in your bucket based on **time** (age of the file).

Why it's important: It helps you **save cost** and **keep your bucket clean** — by moving old files to cheaper storage classes or deleting unneeded data.

Replication rule configuration

Replication rule name

Enter rule ID

Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.

Status

Choose whether the rule will be enabled or disabled when created.

- ☒ Enabled
- ☐ Disabled

Priority

The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.

0

Source bucket

Source bucket name

mentore-solution001

Source Region

Asia Pacific (Mumbai) ap-south-1

Choose a rule scope

- ☒ Limit the scope of this rule using one or more filters
- ☐ Apply to all objects in the bucket

Filter type

You can filter objects by prefix, object tags, or a combination of both.

Replication Rules

What it is: Replication automatically **copies objects** from one bucket to another, either in the **same region** or in a **different region**.

Why it's important: It helps with **backup**, **disaster recovery**, and **data availability** in multiple regions.

◆ Types of Replication:

1. Cross-Region Replication (CRR):

- Copies data from one region to another.
- Example: From Mumbai (ap-south-1) to Singapore (ap-southeast-1).
- Helps with disaster recovery. 🌐

2. Same-Region Replication (SRR):

- Copies data within the same region, but to another bucket.
- Example: For compliance or internal backups.

How It Works:

1. You enable **versioning** in both source and destination buckets.
2. Create a **replication rule** in the Management tab.
3. Define:
 - Source bucket
 - Destination bucket
 - What objects to replicate (prefix, tag, or all)
4. AWS automatically copies every new object that matches the rule.

